CASE 32
DEFICIENCY OF THE C8 COMPLEMENT COMPONENT

A loss of the lytic function of complement.

The complement system of plasma proteins is an effector mechanism of both innate and adaptive immunity that tags pathogens for destruction (see Case 31). The assembly of the so-called terminal components of the complement system (C5 through C9; Fig. 32.1) on the surface of a bacterial cell or a human cell results in the formation of protein complexes that make pores in the cell membrane, leading to cell lysis and death.

Assembly of the terminal complex is initiated when the fifth component of complement, C5, binds to the cell surface and is cleaved by C5 convertase. This reaction releases C5a, a peptide with potent chemotactic activity, from the α, or heavy chain, of C5. The rest of the molecule, C5b, binds C6 to initiate the formation of the so-called membrane-attack complex of terminal complement components. The C5b6 complex binds one C7 molecule, resulting in the exposure of a hydrophobic site on C7 that enables the complex to sink partway into the lipid bilayer of a cell membrane. Complement component C8, unlike the single-chain C6 and C7, is composed of three chains, C8α, C8β, and C8γ. The C8γ chain binds to the C5a67 complex and enables the hydrophobic portion of C8 to embed itself in the cell membrane (Fig. 32.2). This last event induces the polymerization of 10–16 molecules of C9 to make a cylindrical structure, the membrane-attack complex, which forms a pore in the cell membrane. The diameter of the pore is approximately 100 Å, and through this channel sodium and water enter the cell, which swells until it bursts (lysis). The cell-surface protein CD59, which is found in most mammalian cell membranes, inhibits the action of C8 on C9, thereby preventing formation of the membrane-attack complex on the cells of the body but not on those of bacteria.

Curiously, humans with a genetic deficiency in C5, C6, C7, C8, or C9 have an increased susceptibility to systemically invasive infection with bacteria of the genus *Neisseria*. Two common pathogens, *N. meningitidis* and *N. gonorrhoeae*, belong to this genus; the former causes endemic and epidemic meningitis, and the latter the sexually transmitted disease gonorrhea.

TOPICS BEARING ON THIS CASE:

Terminal complement components

Classical pathway of complement activation

The terminal complement components that form the membrane-attack complex		
Native protein	Active component	Function
C5	C5a	Small peptide mediator of inflammation (high activity)
	C5b	Initiates assembly of the membrane-attack system
C6	C6	Binds C5b; forms acceptor for C7
C7	C7	Binds C5b6; amphiphilic complex inserts in lipid bilayer
C8	C8	Binds C5b67; initiates C9 polymerization
C9	$C9_n$	Polymerizes to C5b678 to form a membrane-spanning channel, lysing cell

The case of Dolly Oblonsky: recurrent meningococcal infection leads to the discovery of a complement deficiency.

Dolly Oblonsky was doing well in her first year at university when she developed a cough and diarrhea. She felt very tired and achy and went to bed early in her dormitory room. The next morning she woke early with a severe headache. She felt sick and her neck seemed to be stiff. Her roommate took her to the university infirmary emergency room. She seemed ill and somewhat confused to the nurse on duty, who found that Dolly had a blood pressure of 70/40 (low), a pulse of 124 (fast), a respiratory rate of 24 (increased), and a temperature of 39.2°C (elevated). The nurse noticed a petechial rash (small areas of reddish-purple discoloration) on Dolly's chest and urgently summoned the physician on call, Dr Tolstoy. He found that Dolly had a red throat with moderate enlargement of the tonsils. No other physical symptoms were apparent except for the neck stiffness and the petechial rash on her palate, trunk, and extremities. Dr Tolstoy immediately obtained blood for blood counts and bacterial cultures, and started intravenous administration of the antibiotic Cephtriaxone because he suspected meningococcal meningitis. He then performed a lumbar puncture to obtain cerebrospinal fluid (CSF).

The blood cultures grew *Neisseria meningitidis* (meningococcus) serogroup C. The CSF proved sterile but contained 20 white blood cells μl^{-1} (an abnormally high number; CSF usually contains five or fewer white blood cells μl^{-1}). Dolly's hematocrit was 36.5% (slightly low), and her blood white-cell count was 8700 cells μl^{-1}, of which 90% were neutrophils and 10% were lymphocytes.

Dolly quickly improved on intravenous antibiotic therapy. The fever disappeared, she became alert, and her neck stiffness resolved over the next 72 hours. Blood taken after 24 hours of antibiotic treatment proved sterile on culture, and she was discharged from the infirmary.

Dolly told Dr Tolstoy that she had had meningococcal meningitis in her third year of high school. She was told at the time that she had positive CSF and blood cultures for *N. meningitidis* serogroup Y. Dr Tolstoy suspected that Dolly might have a complement deficiency and sent a blood sample for a CH_{50} assay, which tests the ability of the blood to perform complement-mediated hemolysis. A week later he received the report: Dolly's CH_{50} was zero, which means that her serum contained no hemolytic complement activity.

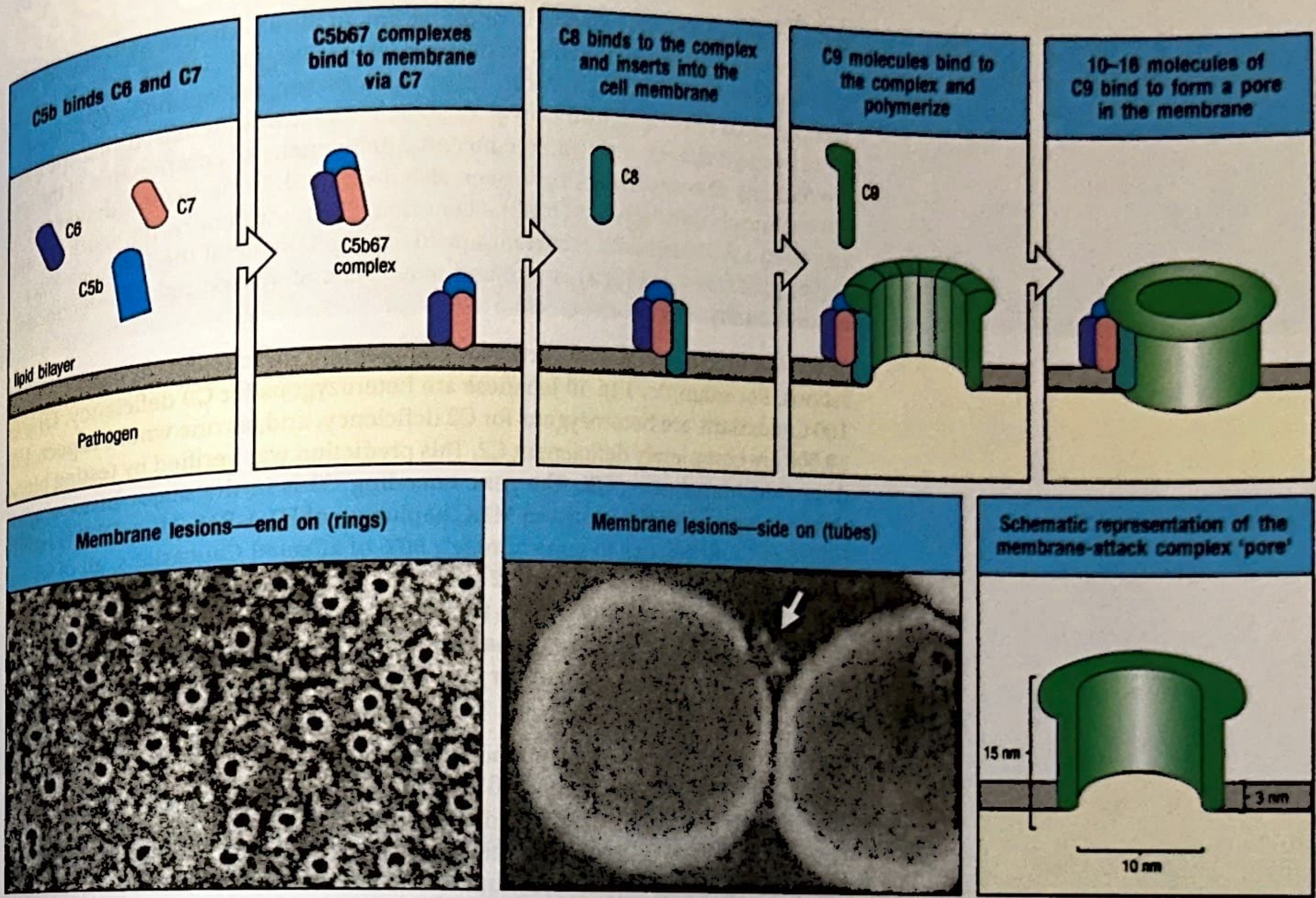

Fig. 32.2 **Assembly of the membrane-attack complex generates a pore in the lipid bilayer membrane.** The sequence of steps and their approximate appearance are shown here in schematic form. C5b triggers the assembly of a complex of one molecule each of C6, C7, and C8, in that order. C7 and C8 undergo conformational changes that expose hydrophobic domains that insert into the membrane. This complex causes moderate membrane damage in its own right, and also serves to induce the polymerization of C9, again with the exposure of a hydrophobic site. Up to 16 molecules of C9 are then added to the assembly to generate a channel 100 Å in diameter in the membrane. This channel disrupts the bacterial cell membrane, killing the bacterium. The electron micrographs show erythrocyte membranes with membrane-attack complexes in two orientations, end on and side on. Photographs courtesy of S. Bhakdi and J. Tranum-Jensen.

Dolly was one of five sisters. Blood samples were obtained from all of them to assay their CH_{50}, which turned out also to be zero in her three younger sisters, ages 9, 14, and 17. The two youngest sisters were well; the 17-year-old sister had had meningococcal meningitis the year before. An older sister, age 20, had a normal CH_{50}. The father was unavailable for testing, but the mother proved to have a half-normal CH_{50}. The sera of all the sisters and the mother were tested in a double-diffusion assay in agar to ascertain whether C5, C6, C7, C8, and C9 were present in their blood. Dolly's serum lacked C8, as did that of her three sisters with CH_{50} values of zero. Her mother had a half-normal level of C8. Her older sister had no evident deficiency of these complement components. The entire family was given tetravalent meningococcal vaccine.

Complement component deficiencies.

Inherited deficiencies of all the components of the classical and alternative pathways have been found in humans. Deficiencies of C1q, C1r, C1s, C4, or C2 of the classical pathway result in immune-complex disease, which frequently proves fatal. Inherited defects in factor D, or properdin, of the alternative pathway, as well as defects in components of the membrane-attack complex, C5, C6, C7, C8, or C9, on which both pathways converge, result in increased susceptibility to neisserial infections, as illustrated by this case. C3 deficiency causes increased susceptibility to all pyogenic infections. Finally, congenital deficiencies of components of the lectin pathway of complement have been also described. In particular, deficiency of the mannose-binding lectin (MBL) is common (5% in the general population), and has been associated with recurrent infections. Deficiency of the MBL-associated serine protease-2 (MASP2) is rare and may cause increased risk of infections or autoimmunity.

There is a high frequency of particular complement deficiencies in certain populations. For example, 1 in 40 Japanese are heterozygous for C9 deficiency. One in 100 Caucasians are heterozygous for C2 deficiency, and, as one would expect, 1 in 10,000 are completely deficient in C2. This prediction was verified by testing blood donors in Manchester, UK. The gene encoding C2 is in the major histocompatibility complex (MHC) locus. An MHC haplotype of HLA-B18, HLA-DR2 is tightly linked to C2 deficiency in approximately 90% of affected Caucasians, all of whom carry the same mutation in the C2 gene. The genes encoding C4 are also in the MHC complex. There are two genes for C4—*C4A* and *C4B*—that are present in variable numbers of copies in the human population. Thirty percent of humans fail to express the products of one, two, or three of these genes, and thus may have a low C4 titer in their blood.

Deficiency of C8 in Caucasian populations is caused by a few well-known mutations in the gene encoding the C8β chain. C8 deficiency in African Americans in the USA, in contrast, is commonly due to mutations in the gene encoding the C8α chain. More than 90% of the mutations in C8β are due to C→T transitions in the *C8B* gene encoding C8β. Such a mutation was carried by Dolly and her sisters and was presumably inherited from her Caucasian father, who was not available for testing. The mother was of Filipino origin and was heterozygous for another type of mutation in the *C8B* gene not found in Caucasians—hence her half-normal levels of C8. Dolly and her three affected sisters thus have a compound heterozygous deficiency for the *C8B* gene.

Questions.

1 *Hemolytic complement levels were determined by measuring the CH_{50}. What is that, and how is the test performed?*

2 *Bacteria of the genus* Neisseria *are encapsulated in a thick carbohydrate capsule. Clinical evidence shows that the membrane-attack complex is vital in host defense against these bacteria. It is obvious that the hydrophobic components of the membrane-attack complex cannot penetrate the polysaccharides of the bacterial capsule. How, then, is complement involved in bacterial killing?*

3 *Patients who are completely deficient in C1q, C1r, C1s, or C4 sustain persistent and severe immune-complex disease such as glomerulonephritis, which can ultimately be fatal. In contrast, patients who are C2-deficient usually sustain only minor forms of immune-complex disease. How do you explain this difference?*